

MRUNAL MOHAN VIBHUTE

Gainesville FL, 32608 | 3522191468 | mvibhute@ufl.edu | [linkedin.com/in/mrunal-vibhute-b6b392235](https://www.linkedin.com/in/mrunal-vibhute-b6b392235) | github.com/Mrun9

Education

University of Florida

M.S., Artificial Intelligence Systems (GPA: 3.89/4.0)

- Coursework: AI Systems, ML for AI Systems, Trustworthy ML, AI Ethics, Image Processing and Computer Vision, Applied Deep Learning

Aug 2025 - May 2027

Gainesville, FL

Cummins College of Engineering

B.Tech, Computer Engineering (GPA: 8.7/10)

- Achievements: Honors: Data Science & ML
- Coursework: Database Management Systems, Software Engineering, Data Structures & Algorithms, Cloud Computing, DevOps, Computer Networks, Project Management, Business Intelligence, Statistics, Android Development, Operating Systems

Aug 2021 - May 2025

Pune, India

Experience

SERMOS Lab | Software & ML Research Assistant

Jan 2026 - Present

- Develop scalable preprocessing pipelines and a foundational ML model to process large geospatial and traffic datasets, enabling more accurate wildfire forecasting.
- Support experimentation and deployment of AI forecasting models and improving evacuation-model accuracy for emergency planners.

Center for Research Computing | Research Intern

Jun 2024 - Dec 2024

- Developed an AI agent using Selenium and LLM agents to collect structured data from **200+ webpages** in average of **92 minutes** with automated CAPTCHA solving and structured CSV export.
- Benchmarked **8 different LLM models** on their ability of CAPTCHA-solving. Best accuracy obtained is **93%**.
- Improved processing performance and reliability by applying computer-vision preprocessing (OpenCV) and streamlining inference workflows, which reduced processing time and lowered error rates

Projects

SearchMind AI | Python, LangChain, LlamaIndex, ChromaDB, FastAPI, Open-Source LLMs

Apr 2026 - Present

- Developed an agentic AI copilot capable of transforming raw multimodal data into testable hypotheses using retrieval-augmented generation and tool-calling LLM workflows.
- Built semantic indexing and memory pipelines with vector embeddings to enable contextual reasoning and long-document interaction.
- Implemented autonomous evidence validation and analytical reasoning modules using Python, LangChain, and open-source LLM ecosystems.

Adversarial Jailbreak Prompt Detection for LLMs | Python, JavaScript, Docker, APIs

Jan 2026 - May 2026

- Built a browser-based monitoring tool for real-time classification of adversarial LLM prompts with **89% accuracy** using a hybrid architecture of DistilBERT and a NLP based mutator module for creating harmful jailbreak prompt variants.
- Developed **scalable semantic retrieval cache** and classification workflows using vector search.
- Optimized inference **latency by 30%** and improved monitoring reliability for live prompt analysis.

Co-Wingman: Real-Time AI Speech Anxiety Detection | Python, PyTorch, Whisper, SHAP, Docker

Aug 2025 - Nov 2025

- Developed a real-time speech anxiety detection pipeline using CNN-LSTM models on MFCC audio features, achieving **96% recall with 1.2–1.5s inference latency** on rolling 5-second windows.
- Integrated Whisper transcription, MiniLM keyword extraction, and SHAP explainability for interpretable stress analysis during live conversations.
- Deployed the system using Docker with Prometheus and Grafana monitoring for latency tracking, inference reliability, and confidence drift detection.

CruxCapture: Multimodal Summarization Platform | Java, Android Studio, Firebase, REST API

Jan 2024 - Apr 2024

- Developed an Android application that summarizes text, audio, and video content using external AI APIs with latency of **5 seconds**.
- Implemented Firebase authentication, cloud synchronization, and user dashboard functionality.
- Added PDF export and content-sharing workflows using email and messaging integrations.

Publications

- **Multimodal Web Agents for Automated (Dark) Web Navigation.** Proposes an autonomous browsing system that uses visual and textual cues to traverse both surface and hidden websites for complex information-gathering tasks. **ICAART 2025 (DOI: 10.5220/0013171600003890)**
- **Interpretable Fake News Detection Using Neural Networks and LIME.** Builds a transparent misinformation classifier that highlights human-readable evidence within articles to explain why content is considered unreliable. **ISD4SD 2025 (DOI: 10.1007/978303206662646)**
- **USD-INR Exchange Rate Prediction Using LSTM.** Analyzes macroeconomic indicators and historical market data with a deep recurrent architecture to forecast future movements in a major currency pair. **ICICT 2024 (DOI: 10.1007/978981973559424)**

Technical Skills

- **Languages:** Python, Java, JavaScript, HTML/CSS, C, R, MATLAB
- **Backend & Systems:** Flask, Restful API, Docker, Selenium, Firebase
- **Cloud & Dev Tools:** AWS, GitHub, Postman, VSCode, Linux, Tableau, Grafana
- **Databases & Frameworks:** SQL, Firebase, Streamlit, Gradio
- **Concepts:** Machine Learning, Deep Learning, Object-Oriented programming, API Integration, Distributed Systems